- Attributes about the *subject*; that is, the user making the request. This could include their username, any groups they belong to, how they were authenticated, when they last authenticated, and so on.
- Attributes about the *resource* or object being accessed, such as the URI of the resource or a security label ("TOP SECRET" for example).
- Attributes about the *action* the user is trying to perform, such as the HTTP method.
- Attributes about the *environment* or context in which the operation is taking place. This might include the local time of day, or the location of the user performing the action.

The output of ABAC is then an allow or deny decision, as shown in figure 8.3.

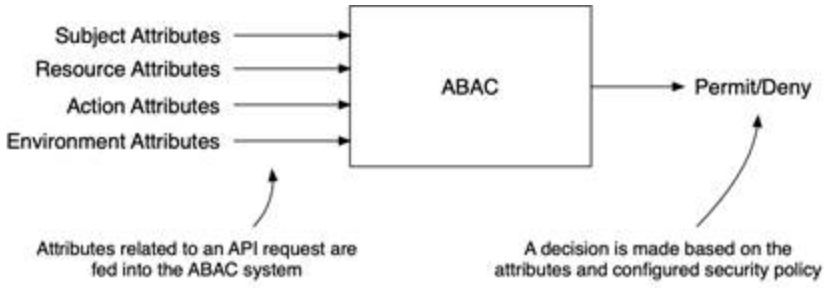


Figure 8.3 In an ABAC system access control decisions are made dynamically based on attributes describing the subject, resource, action, and environment or context of the API request.

Listing 8.9 shows example code for gathering attribute values to feed into an ABAC decision process in the Natter